PROJECT REPORT

Student Performance Analyzer

A Predictive Analytics System with Interactive Web Dashboard

Built using Random Forest Regression, Fast-API, and React

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Students Performance Prediction System

Abstract

The Student Performance Prediction System is a machine learning-based predictive analytics application designed to estimate students' examination scores based on their academic and behavioral attributes. The project utilizes a supervised machine learning approach to identify relationships between study habits and academic performance.

Extra effort:

A web-based dashboard was also developed by me to allow users to input student information and receive real-time predictions.

Problem Statement

Educational institutions often struggle to identify students who may perform poorly before examinations. Early prediction of student performance enables teachers and students to take preventive actions. This project aims to predict examination scores using historical academic data.

Objectives

- Predict students' examination scores using Machine Learning.
- Perform data preprocessing and visualization.
- Evaluate machine learning performance.
- Build an interactive web dashboard. (extra effort)
- Deploy the prediction model using FastAPI through vercel. (extra effort)

Dataset

Source Link: Kaggle

Dataset Name: Student Performance Prediction Dataset **(10,000 Records)**

Features:

- Study Hours
- Attendance Percentage

- Sleep Hours
- Internet Usage
- Assignments Completed
- Previous Scores

Target:

- Exam Score

Technologies Used

- Python
- Google Colab (IDE)
- Scikit-learn
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Joblib
- FastAPI
- React.js
- Vercel (Deployment Frontend + Backend)

Machine Learning Algorithm

- Random Forest Regressor

Machine Learning Workflow

- Dataset Collection
- Data Cleaning
- Feature Selection
- Train-Test Split
- Model Training
- Model Evaluation
- Model Serialization

Evaluation Metrics

- **MAE (Mean Absolute Error)** : 5.84
- **MSE (Mean Squared Error)** : 65.42
- **RMSE (Root Mean Squared Error)** : 8.09

- **R^2 Score (R-Squared – coefficient of Determination)** : 0.72

Web Based Dashboard

The web based dashboard was made using React.js and FastAPI. Web based Dashboard made the project interactive by making it accessible via an interactive UI for non-technical user.

System Architecture:

[React Frontend] → HTTP requests → [FastAPI Backend] → loads → [Trained Random Forest Model (.pkl)]

Key UI Components

- Intake form — captures the six model inputs (study hours, attendance, sleep, internet usage, assignments completed, previous score)
- Gauge visualization — a semicircular meter with colour-coded risk zones (red / amber / sage) showing the predicted score at a glance
- Habit breakdown bars — each input plotted against a healthy target range, colour-coded to flag which habits need attention
- Recommendations panel ("Teacher's Notes") — dynamically generated text advice based on which specific inputs fall outside healthy ranges
- Placement badge — automatically labels the student "Placement Ready" or "Needs Improvement" based on a 70-point threshold

Conclusion

The project successfully predicts students' examination scores using Machine Learning techniques. It demonstrates the complete machine learning lifecycle, including data preprocessing, model development, deployment, and frontend integration.

Links

- Dashboard
- Github (Python ML Model)
- Google Colab
- Kaggle Dataset
- Github (Dashboard UI)
- Youtube Video